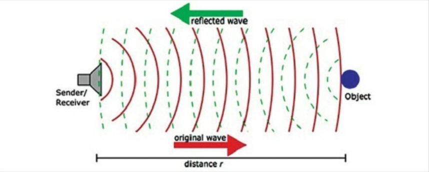

Fig. 2: How microwave motion sensors work

Rajath explains, "Microwave sensors offer a distinct advantage over PIR sensors with a continuous field-of-detection zone. PIR sensors have a slotted detection zone. Therefore PIR sensors may miss out objects in their detection zone." This makes microwaves a better proposition for security.

However, microwaves have higher false alarm rates. Moreover, these are slightly costlier than PIR sensors and consume more energy. Rajath explains, "Microwave sensors typically consume 30 per cent more energy than PIRs. While PIR sensors consume 0.8-1.0W electricity, microwaves take 1.1-1.5W. Microwave sensors may start from ₹ 600 and go up to ₹ 1000. These cost 20 per cent more than PIR sensors."

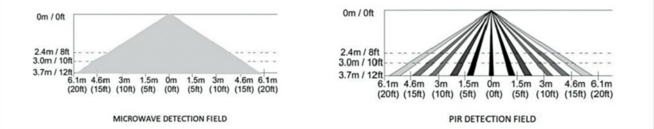

Fig. 3: Technology difference between microwave and PIR sensors

ability. Many industries are installing microwave sensors to monitor the functions of bucket elevators and belt conveyors.

Wadeco, a Japan-headquartered industrial microwave sensor manufacturer, provides microwave sensors for crane proximity detection. These sensors are used to ensure a safe distance between two cranes working on the same line, or between a crane and a building or landing station close-by. City municipalities use microwave sensors to monitor wastewater and sewage levels in sewage lines.

Even medical science is leveraging the low-power properties of microwave sensors. A lot of R&D is underway to come up with non-invasive treatments in medical applications like breast cancer treatment, separation of red blood cells from white blood cells and liver tissue disease detection using microwave sensors.

Challenges with microwave sensors

Microwave sensors have a large coverage range, and their waves penetrating through walls or glass also poses challenges such as false alarms. For instance, when used for lighting control in small office floors, the sensors may switch on the lights even if someone is outside the premises. The high sensitivity of microwave sensors will also detect non-human presence, like animals or some fast-moving objects in the air. On the other hand, microwave sensors cannot reach beyond any metal obstruction.

Are PIR sensors better?

Often, passive infrared (PIR) sensors are used as an alternative to microwave sensors. However, their performance differs owing to the underlying technologies.

As mentioned earlier, microwave sensors have a higher sensitivity and coverage range than PIR sensors. Changlani explains, "Since PIRs read infrared heat signatures in a room, these are not very sensitive if the room itself is warm. Therefore, in warm countries like India, PIR sensors at times may not be able to detect a person, especially in summers. On the other hand, during winters, these sensors become highly sensitive."

Snoozing is another challenge with PIR sensors. Even on occupied floors, PIR sensors may turn off even if there is very little movement. A little body movement instantly turns these on again.

Moreover, thieves may find it easier to fool the PIR detection range.

Overall, PIR sensors are good for electrical applications in smaller and compact premises. Microwave sensors are suited for large-area security applications.

The best results can be achieved by installing both the technologies in a complementary fashion. For example, California's North Bay Vaca Valley hospital invested in a lighting setup transformation for their 225 parking spaces in 2014. The investment included 50 induction luminaires with LED fixtures. PIR sensors and long-range microwave sensors were installed in dedicated zones to save electricity when these areas were not in use. Smart controls were also put in place. The annual audit of this setup reported 14,600-kilowatt-hour reduction in energy consumption, which cumulated to 65 per cent electricity savings. There were annual savings worth $2300 with the help of these sensors complementing each other. EFY